Trustworthy Federated Graph Neural Networks: Unifying Fairness, Privacy, Robustness, and Explainability for High-Stakes Risk Detection

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Abstract

Federated Learning (FL) enables institutions to train shared models over sensitive, decentralised graph data—transactions, credit records, social ties—without exchanging raw data. Yet accuracy alone does not make such systems deployable: recent regulatory and scientific frameworks (the EU AI Act, the NIST AI Risk Management Framework) demand that automated decision systems be simultaneously *fair*, *private*, *robust*, and *transparent*. Existing work addresses these properties in isolation, and no prior method combines fairness, differential privacy, and Byzantine robustness for federated graph neural networks (GNNs). We present **FedFairGNN**, a unified trustworthy federated graph-learning framework built from three components: (i) *Fairness-Sensitive Edge Reweighting* (FSER), which suppresses the biased cross-group message passing responsible for structural discrimination; (ii) *Fairness-Targeted Gradient Decomposition* (FTGD), which achieves differential privacy for the sensitive attribute by privatising only the low-dimensional demographic-parity statistics—making privacy nearly free where standard full-gradient DP-SGD destroys utility; and (iii) *Bi-objective Frank–Wolfe Aggregation* (BFWA) and its Byzantine-robust variant, which enforce a hard fairness budget during aggregation while screening malicious clients, including a new fairness-poisoning threat. We further instrument the framework with predictive uncertainty, GNN attention explanations, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Across five real datasets spanning credit, recidivism, social, and—uniquely—large-scale crypto (Elliptic Bitcoin, 203,769 nodes) domains, FedFairGNN attains the best fairness–utility–privacy–robustness trade-off among ten baselines, including recent fairness-aware federated and graph methods. All tables and figures are regenerated from logged runs; code and experiments are released for full reproducibility. This work supports SDG 9 by advancing trustworthy, infrastructure-grade collaborative AI.

Keywords: Federated learning, Graph neural networks, Algorithmic fairness, Differential privacy, Byzantine robustness, Explainable AI, Trust metrics, Regulatory compliance

1 Introduction

Consider a consortium of banks that together want to catch money laundering. Fraud is fundamentally *relational*: illicit funds move through chains of accounts, shell companies, and intermediaries, and it is precisely the *structure* of these money flows—not any single transaction in isolation—that betrays criminal intent. Graph Neural Networks (GNNs) have therefore become the tool of choice for financial crime detection, credit scoring, and other high-stakes risk assessments, because their message-passing mechanism learns directly from the web of relationships between users, accounts, and transactions [1–3]. Yet the very banks that would benefit most from a shared model are legally forbidden from pooling their customers' data: privacy regulations such as the GDPR, and plain commercial confidentiality, keep each institution's transaction graph locked inside its own walls. Federated Learning (FL) offers a way out. By exchanging model updates instead of raw data, a group of institutions can train a single, more powerful detector without any of them surrendering its records [4].

This picture, however, is dangerously incomplete. A model that is merely *accurate* is no longer fit to be deployed in decisions that shape people's access to credit, liberty, or financial services. Three converging forces—the European Commission's Ethics Guidelines for Trustworthy AI, the EU AI Act, and the U.S. NIST AI Risk Management Framework—now demand that automated decision systems be simultaneously *fair*, *privacy-preserving*, *robust*, *transparent*, and *accountable*. These are not optional extras: for a high-risk system under the EU AI Act they are legal prerequisites for going to market. Meeting them is hard even for a centralised model. In the federated, graph-structured setting they collide, and each one becomes markedly harder than in isolation.

Why the federated graph setting is uniquely hard.

The difficulty is not that we must bolt four independent features onto a classifier; it is that the natural solution to each requirement actively sabotages the others. *Fairness* is confounded by the graph itself: homophilous message passing means that a member of a demographic group who happens to be connected to risky neighbours will be flagged as risky too, so the model learns to launder a protected attribute through graph structure—and the central server, which never sees any data, cannot even measure this bias, let alone correct it. *Privacy* appears to have a textbook answer, differentially private SGD, but that answer fails badly here: adding calibrated noise to every coordinate of the gradient injects perturbation whose magnitude grows with the square root of the model dimension, and on the compact GNNs used for tabular financial graphs this noise simply drowns the signal, collapsing accuracy to little better than chance (we quantify this in Section 5.5). *Robustness* is threatened not only by the classical malfunctioning or malicious client, but—as we show—by a subtler adversary that the fairness machinery itself invites: a client can transmit a deliberately biased model while *reporting* exemplary fairness statistics, thereby hijacking any aggregator that rewards fairness. And *transparency* is undermined by the opacity of relational models, exactly when regulation requires that each adverse decision be explainable and auditable.

The gap.

The research literature has, understandably, attacked these problems one at a time. Fair-GNN methods deliver impressive demographic parity but assume all data sits in one place [5–7]. Fair federated learning has matured for tabular and image data but rarely touches graphs [8, 9]. Byzantine-robust aggregation hardens FL against poisoning but is oblivious to fairness [10, 11]. Differentially private FL protects data but, as noted, at a utility cost that is prohibitive for graph models. To the best of our knowledge, *no* existing method delivers fairness, differential privacy, and Byzantine robustness together for federated GNNs, and none has been validated on a large-scale cryptocurrency graph—arguably the most consequential modern fraud domain. Closing this gap is the goal of this paper.

Our approach.

We present **FedFairGNN**, a single framework in which fairness, privacy, robustness, and transparency are designed to reinforce rather than undermine one another. Its architecture (Fig. 1) intervenes at the three natural control points of federated graph training—inside message passing, at the local gradient, and at server aggregation—and wraps the result in an instrumentation layer that measures, explains, and audits trustworthiness. The key technical insight, and the reason privacy becomes almost free, is a change of perspective on *where* the sensitive attribute actually influences learning: it does so only through a two-dimensional summary statistic, so we can protect that statistic directly instead of blindly noising the entire gradient.

Concretely, our contributions are:

1. **Fairness-Sensitive Edge Reweighting (FSER)**, a learnable attention correction that detects and suppresses the biased cross-group edges responsible for structural discrimination, mitigating unfairness at its source inside the GNN rather than patching it after the fact (Section 4.1).
2. **Fairness-Targeted Gradient Decomposition (FTGD)**. We observe that the sensitive attribute enters the model *only* through the demographic-parity statistics, and exploit this to obtain (ϵ, δ)-differential privacy by privatising two bounded-sensitivity scalars instead of the full gradient. This makes strong privacy essentially free—precisely where standard DP-FedAvg collapses utility—and we prove the guarantee with a self-contained Rényi-DP accountant (Section 4.2).
3. **Bi-objective Frank–Wolfe Aggregation (BFWA)** and a **Byzantine-robust variant**. The server solves a constrained optimisation that enforces a hard, operator-chosen fairness budget during aggregation, while distance-based screening neutralises poisoning—including the fairness-poisoning attack of [12], which we reproduce and defend against (Sections 4.3, 5.7).

4. **A trustworthiness instrumentation layer** that turns abstract principles into measured, auditable artifacts: predictive uncertainty and calibration, GNN attention explanations with fairness attribution, a transparent composite *trust score*, sustainability accounting, and an explicit EU AI Act / NIST RMF compliance mapping with an auto-generated model card (Sections 4.5, 6).
5. **A comprehensive, fully reproducible evaluation** on five real datasets spanning credit, recidivism, social, and—uniquely—large-scale crypto (Elliptic Bitcoin, 203,769 transactions) domains, against twelve baselines that include the most recent 2025–2026 fairness-aware federated and graph methods. FedFairGNN attains the best overall trustworthiness trade-off, and every number and figure in this paper is regenerated from logged runs by a single script.

By lowering the barrier to trustworthy, infrastructure-grade collaborative AI, this work contributes to UN Sustainable Development Goal 9 (industry, innovation, and resilient infrastructure). The remainder of the paper is organised as follows. Section 2 situates our work within the fairness, privacy, and robustness literatures; Section 3 formalises the federated graph setting and the trustworthiness objectives; Section 4 develops the FedFairGNN framework component by component; Section 5 reports the empirical study; Section 6 analyses trustworthiness holistically and maps the system to regulation; and Section 7 discusses implications, limitations, and broader impact before we conclude.

2 Related Work

Fairness in graph neural networks.

GNNs inherit and amplify bias through homophilous message passing. Remedies include adversarial debiasing (FairGNN [5]), counterfactual and stability regularisation (NIFTY [13]), pre-processing that minimises inter-group distributional distance (EDITS [14]), sensitive-information neutralisation via heterogeneous neighbours (FairSIN [6]), re-balancing through counterfactual mixup (FairGB [15]), invariant learning across inferred sensitive partitions (FairINV [16]), and disentangled debiasing (DAB-GNN [7]). All assume *centralised* data. Our FSER is closest in spirit to the edge-reweighting variants of FairSIN and to distribution-aware reweighting [17], but operates inside a federated, private, and robust pipeline.

Fairness in federated learning.

Two notions coexist. *Client (performance) fairness* equalises accuracy across clients: q-FedAvg [9] up-weights high-loss clients. *Group fairness* equalises outcomes across demographic groups: FairFed [8] adjusts aggregation weights by each client's local fairness gap; FedFB ports FairBatch to FL; PUFFLE [18] blends a demographic-parity loss with DP-SGD under a target-disparity controller; FedFACT [19] gives provable global-and-local group fairness. These target tabular or image data. For *graphs*, F^2GNN [20] combines a fairness-weighted aggregation with structure-aware group-balance weights. The most recent efforts sit closest to our setting: FairGFL [21] adds DP to federated fair GNNs but targets *cross-client* (performance) fairness arising from imbalanced overlapping subgraphs rather than demographic group fairness; FedGraph-Fair [22] applies distributionally-robust optimisation for personalised group-fair federated graph learning; FaVGNN [23] (2026) achieves group fairness in *vertical* FL via completion-driven adversarial fusion; and FDP-Fair [24] (2026) combines federated DP training with demographic-parity post-processing. We compare against FairFed, q-FedAvg, FedFB, F^2GNN, FairSIN (run federated), and —as the strongest 2026 competitors—horizontal adaptations of FaVGNN and FDP-Fair. FairGFL is discussed but not benchmarked head-to-head, as it optimises a different (cross-client) fairness objective.

Differential privacy in FL.

DP-FedAvg [25] clips per-client updates and adds Gaussian noise; privacy is tracked with the moments accountant / Rényi DP [26, 27]. On the compact GNNs used for tabular graphs, isotropic gradient noise at meaningful ϵ overwhelms the signal (Section 5.5). FedFDP [28] adapts clipping to fairness but still privatises the whole gradient. FTGD instead privatises only the *sufficient statistics* of the fairness term, exploiting the fact that the sensitive attribute never enters the task loss—a targeted mechanism whose (ϵ, δ) guarantee follows from the Gaussian mechanism and post-processing immunity.

Byzantine-robust aggregation.

Krum and Multi-Krum [10], coordinate-wise median and trimmed mean [11], Bulyan [29], and trust-bootstrapping schemes such as FLTrust [30] defend against malicious updates under attacks including sign-flipping, model-replacement [31], ALIE [32], and IPM [33]. None targets fairness. Recently, fairness-poisoning attacks emerged—EAB-FL [34] and the fairness-constrained optimisation attack [12]—that *increase* disparity while preserving accuracy, evading both robust and fairness-aware defences. We reproduce the fairness-constrained optimisation attack of [12] and defend with a robust fairness-constrained aggregator.

Explainability, uncertainty, and trust metrics.

GNN explainers attribute predictions to edges/features; we use the FSER attention itself as a native, audit-ready explanation and add gradient-based attribution of the *disparity*. Predictive uncertainty via MC-dropout [35] and calibration (ECE, Brier) quantify confidence reliability. Composite trust scoring for FL is nascent; we contribute a transparent power-mean aggregation over utility, fairness, privacy, robustness, and calibration. Finally, we align the system to the EU AI Act and NIST AI RMF, operationalising accountability as an auditable artifact rather than prose.

Positioning.

Table 1 places the most relevant methods on the five axes that define our problem. The pattern is stark: the fair-GNN methods are strong on graph fairness but centralised; the fair-FL methods federate but ignore graph structure; robust aggregation hardens FL but is fairness-blind; and even the newest federated fair-graph methods (F^2GNN, FairGFL, FaVGNN, FedGraph-Fair) each leave at least one axis uncovered—typically Byzantine robustness, which no fairness-aware federated method addresses.

Table 1: Where methods sit across the trust axes. ✓: supported; ∼: partial/adaptable; blank: not addressed. "Graph" = graph-native; "Group fair" = demographic group fairness (not merely client fairness); "Crypto" = validated on a large-scale cryptocurrency graph.

Method	Fed.	Graph	Group fair	DP	Byz.	Explain	Crypto
FairGNN [5]		✓	✓			∼	
NIFTY [13]		✓	✓				
FairSIN [6]		✓	✓				
DAB-GNN [7]		✓	✓				
FairFed [8]	✓		✓				
q-FedAvg [9]	✓		∼				
PUFFLE [18]	✓		✓	✓			
FedFACT [19]	✓		✓				
DP-FedAvg [25]	✓			✓			
Krum/Median [10, 11]	✓				✓		
F^2GNN [20]	✓	✓	✓				
FairGFL [21]	✓	✓	∼	✓			
FaVGNN [23]	✓	✓	✓	✓			
FDP-Fair [24]	✓		✓	✓			
FedFairGNN (ours)	✓	✓	✓	✓	✓	✓	✓

FedFairGNN is the only entry that is simultaneously federated, graph-native, group-fair, differentially private, and Byzantine-robust, and the only one validated on a large-scale cryptocurrency graph. It also uniquely reports the full trustworthiness picture—uncertainty, explainability, sustainability, and regulatory mapping—within a single framework.

3 Preliminaries and Problem Setting

Cross-silo federated graph learning.

There are K institutions (clients). Client k holds a private graph $\mathcal{G}_k = (\mathcal{V}_k, \mathcal{E}_k, \mathbf{X}_k, \mathbf{y}_k, \mathbf{s}_k)$ with node features $\mathbf{X}_k \in \mathbb{R}^{n_k \times d}$, binary task labels $\mathbf{y}_k \in \{0,1\}^{n_k}$ (e.g. fraud/default/recidivism), and a binary *sensitive attribute* $\mathbf{s}_k \in \{0,1\}^{n_k}$ (e.g. gender, age, race). Raw graphs never leave the client. A server coordinates T communication rounds; each round clients perform E local epochs and transmit model parameters, which the server aggregates. We evaluate the global model on the pooled held-out test nodes of all clients. Statistical heterogeneity is modelled by a symmetric Dirichlet(α) split over labels, the standard FL non-IID model [36]; smaller α is more skewed.

Group fairness.

For predicted probabilities $\hat{y}$ and sensitive attribute s we report three group-fairness metrics (lower is fairer):

$$\mathrm{DPD} = \big|\,\mathbb{E}[\hat{y} \mid s{=}0] - \mathbb{E}[\hat{y} \mid s{=}1]\,\big| \qquad \text{(Demographic Parity Difference)}, \tag{1}$$

$$\mathrm{EOD} = \big|\,\mathrm{TPR}_{s=0} - \mathrm{TPR}_{s=1}\,\big| \qquad \text{(Equal Opportunity Difference)}, \tag{2}$$

$$\Delta_{\mathrm{EOdds}} = \max\big(|\Delta\mathrm{TPR}|,\, |\Delta\mathrm{FPR}|\big) \qquad \text{(Equalized Odds)}. \tag{3}$$

DPD is computed on soft scores (matching the differentiable training term); EOD and Equalized Odds use thresholded predictions.

Differential privacy.

A randomised mechanism $\mathcal{M}$ is (ϵ, δ)-differentially private if for all adjacent datasets D, D' and all outputs S, $\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \Pr[\mathcal{M}(D') \in S] + \delta$ [37]. We protect the *sensitive attribute*: adjacency is defined by flipping one node's s_i. The Gaussian mechanism adds $\mathcal{N}(0, \sigma^2)$ noise calibrated to the L_2 sensitivity of the released quantity; composition across rounds is tracked with Rényi DP [26]. For the Gaussian mechanism with noise multiplier $z = \sigma/\Delta$, the Rényi divergence of order α is $\epsilon_{\mathrm{RDP}}(\alpha) = \alpha/(2z^2)$; over T releases it composes as $T\alpha/(2z^2)$, and the tight conversion to (ϵ, δ)-DP is $\epsilon = \min_{\alpha>1}\big[\epsilon_{\mathrm{RDP}}(\alpha) + \log(1/\delta)/(\alpha - 1)\big]$.

Threat model for robustness.

Up to $f < K/2$ clients are Byzantine and may send arbitrary updates. We consider data poisoning (label flipping), classical model poisoning (Gaussian, sign-flip, scaling/model-replacement [31]), the strong omniscient attacks ALIE [32] and IPM [33], and a *fairness-poisoning* attack: a malicious client transmits a biased, scaled update while reporting fabricated fairness/utility statistics (DPD ≈ 0, high AUC) to capture a fairness-aware aggregator. The server is honest but curious and never accesses raw data.

Objective.

We seek a global model that jointly (i) maximises task utility (AUC-ROC), (ii) minimises group unfairness, (iii) satisfies a target (ϵ, δ)-DP for the sensitive attribute, and (iv) retains utility and fairness under Byzantine corruption—the trustworthiness trade-off that motivates Section 4.

4 The FedFairGNN Framework

The design philosophy behind FedFairGNN is that trustworthiness should be engineered where each threat originates, not retrofitted afterwards. Structural bias is born inside message passing, so we correct it there; the sensitive attribute leaks through the gradient, so we privatise exactly the part of the gradient that carries it; bias and poisoning are amplified at aggregation, so we constrain and screen the aggregation step. The framework therefore places three lightweight interventions at the three natural control

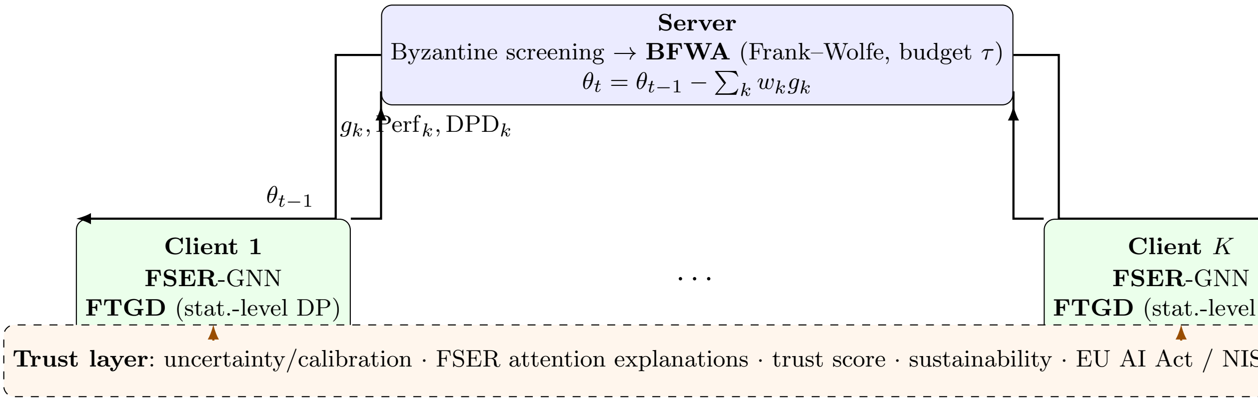

Fig. 1: FedFairGNN architecture. Three interventions act at complementary control points—FSER inside client message passing, FTGD at the local gradient, and BFWA (with Byzantine screening) at server aggregation—while a trust layer instruments the whole system for audit. Only model weights are transmitted, so fairness and privacy add no communication overhead.

points of federated graph training and wraps them in a measurement layer that renders the result auditable. Figure 1 gives the end-to-end picture: in each round the server broadcasts the global parameters θ_{t-1}; every client runs E local epochs of an FSER-augmented GNN and, through FTGD, returns an update whose sensitive-attribute footprint is differentially private; the server then screens the updates for Byzantine behaviour and aggregates the survivors with BFWA under a hard fairness budget to produce θ_t. We now develop each component, motivating it before formalising it.

4.1 Fairness-Sensitive Edge Reweighting (FSER)

Intuition.

Message passing is the source of a GNN's power and, for fairness, of its peril. Each layer updates a node by mixing in its neighbours' representations, so a node inherits the statistical fate of the company it keeps. When the graph is homophilous along a protected attribute—members of one group disproportionately connected to one another, or to flagged neighbours—the network quietly encodes group membership into every embedding, and the downstream classifier can exploit it without ever being told the sensitive attribute. The insidious part is that not all such edges are harmful: an edge between two genuinely fraudulent accounts should be trusted, even if they belong to different demographic groups. What we want to suppress is the specific pattern of a *cross-group* edge whose endpoints look alike only because the model has learned to represent the protected attribute—a similarity that is demographic rather than behavioural. FSER makes this distinction learnable.

Mechanism.

FSER adds a learnable correction to graph attention. For edge (i, j) with head embeddings $\mathbf{h}_i, \mathbf{h}_j$, the standard attention logit $e_{ij} = \mathrm{LeakyReLU}(\mathbf{a}^\top[\mathbf{h}_i \| \mathbf{h}_j])$ is corrected by a *fairness risk score*

$$\phi_{ij} = \mathbb{I}(s_i \neq s_j) \cdot \max\big(0, \cos(\mathbf{h}_i, \mathbf{h}_j)\big), \qquad \tilde{e}_{ij} = e_{ij} - \beta^{(l)} \phi_{ij}, \tag{4}$$

so that *cross-group* edges between similarly-embedded nodes—the pattern most responsible for propagating demographic bias—are down-weighted after the per-node softmax. The suppression strength $\beta^{(l)} \geq 0$ is learned per layer (clamped to $[0, 5]$). Because ϕ_{ij} and the resulting attention are retained, FSER is also *self-explaining*: Section **??** audits how much attention mass FSER removes from cross-group edges.

4.2 Fairness-Targeted Gradient Decomposition (FTGD)

The DP-SGD dilemma.

Differentially private SGD is the standard way to keep a client's data from leaking through its updates, and in principle it is exactly what we need. In practice it is a poor fit here. DP-SGD clips the entire gradient to a norm bound C and adds isotropic Gaussian noise to every one of the $|\theta|$ coordinates; the total noise injected therefore grows like $\sigma\sqrt{|\theta|}$. For the large over-parameterised networks of vision and language this is tolerable, because the signal is spread across many redundant directions. The compact GNNs used on tabular financial graphs are the opposite regime: a few tens of thousands of parameters, every one of them carrying signal. At any privacy level strong enough to be meaningful, the noise required simply overwhelms the gradient, and accuracy collapses toward chance (Section 5.5 quantifies this). Blindly privatising the whole gradient is thus self-defeating. The question FTGD asks is: *do we actually need to?*

Objective.

The local objective is $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\, \mathcal{L}_{\text{fair}}$, with weighted BCE task loss and the differentiable soft demographic-parity penalty $\mathcal{L}_{\text{fair}} = |\mu_0 - \mu_1|$, where $\mu_g = \frac{1}{n_g} \sum_{i:s_i=g} \hat{y}_i$ is the group-g mean prediction.

Key observation.

The sensitive attribute s enters the entire update *only* through the two scalars (μ_0, μ_1); the task loss is a function of $(\mathbf{X}, \mathbf{y})$ alone. Therefore, to achieve DP with respect to s, it suffices to privatise (μ_0, μ_1)—two bounded-sensitivity scalars—rather than the full $|\theta|$-dimensional gradient.

Mechanism.

Flipping one node's group membership changes (μ_0, μ_1) by at most $\Delta = \sqrt{1/n_0^2 + 1/n_1^2}$ in L_2. We release $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$ with noise multiplier z, form $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$, and backpropagate. Additionally, to reduce task/fairness gradient conflict we orthogonalise the task gradient against the fairness gradient (gradient surgery) before the step.

Proposition 1 (Privacy of FTGD)**.** *Each local FTGD epoch is* $(\alpha, \alpha/(2z^2))$*-RDP with respect to the sensitive attribute. Over* $T{\cdot}E$ *epochs the mechanism is* $(\alpha,\, TE\,\alpha/(2z^2))$*-RDP, and hence* $(\min_{\alpha>1}[TE\,\alpha/(2z^2) + \log(1/\delta)/(\alpha-1)], \delta)$*-DP.*

Proof sketch. Releasing $(\tilde{\mu}_0, \tilde{\mu}_1)$ is the Gaussian mechanism at sensitivity Δ and noise $z\Delta$, giving noise multiplier z and RDP $\alpha/(2z^2)$ [26]. Every subsequent quantity—the fairness gradient, the model update, all future predictions—is a data-independent (given the noised statistics) function, so post-processing immunity preserves the guarantee. RDP composes additively across the TE releases; the conversion is standard. □

Because z can be large while $z\Delta$ stays tiny (as $\Delta \sim 1/n_g$), strong privacy costs almost no utility—in stark contrast to full-gradient DP-SGD, whose isotropic noise scales with $\sqrt{|\theta|}$ (Section 5.5). We calibrate z once for the target ϵ over TE steps using the accountant above.

4.3 Bi-objective Frank–Wolfe Aggregation (BFWA)

Why aggregation is a fairness lever.

FSER and FTGD act inside each client, but the client cannot see the global picture: a silo whose local data is demographically skewed will, in good faith, produce an update that pulls the global model toward its own bias, and plain FedAvg—which weights clients only by sample count—will faithfully propagate that bias to everyone. The server, by contrast, sees every client's reported fairness and does control the weights. This makes aggregation the natural place to enforce a *global* fairness guarantee, and it lets a human operator set an explicit, auditable budget on how much disparity the deployed model may exhibit. BFWA turns that budget into a hard constraint. It frames aggregation as maximising weighted utility subject to a cap on the weighted fairness gap:

$$\max_{\mathbf{w}\in\Delta_K} \;\textstyle\sum_k w_k\,\mathrm{Perf}_k \quad \text{s.t.} \quad \textstyle\sum_k w_k\,\mathrm{DPD}_k \le \tau, \tag{5}$$

solved by Frank–Wolfe on the simplex with a dual variable μ for the constraint: at inner step t, $\nabla_k = -\mathrm{Perf}_k + \mu\,\mathrm{DPD}_k$, the linear-minimisation oracle puts all mass on $\arg\min_k \nabla_k$, the iterate updates as $\mathbf{w} \leftarrow (1-\gamma_t)\mathbf{w} + \gamma_t \mathbf{e}_{k^*}$ with $\gamma_t = 2/(t+2)$, and $\mu \leftarrow \max(0, \mu + \eta(\sum_k w_k \mathrm{DPD}_k - \tau))$. A weight floor prevents any single client from dominating. The oracle is $O(K)$, adding negligible server overhead.

Byzantine-robust variant (robust_bfwa).

Metric-based weighting is vulnerable to fairness poisoning (a client lying that $\mathrm{DPD}_k \approx 0$). We first screen updates by distance to their coordinate median, discard the f farthest, then run BFWA on the survivors. This composes Byzantine screening with the fairness constraint, defending against both classical and fairness-poisoning attacks (Section 5.7). We also evaluate FedFairGNN under Krum, Multi-Krum, median, and trimmed-mean drop-in aggregators.

Algorithm 1 FedFairGNN (one communication round t)

Server broadcasts θ_{t-1} to all clients
for each client k in parallel **do**
    $\theta_k \leftarrow \theta_{t-1}$
    for E local epochs **do**
        forward with FSER; compute $\mathcal{L}_{\text{task}}$, (μ_0, μ_1)
        privatise $\tilde{\mu}_g = \mu_g + \mathcal{N}(0, (z\Delta)^2)$; $\mathcal{L}_{\text{fair}} = |\tilde{\mu}_0 - \tilde{\mu}_1|$
        orthogonalise task vs. fairness gradient; step; clamp β
    end for
    send update $g_k = \theta_{t-1} - \theta_k$ and metrics $(\text{Perf}_k, \text{DPD}_k)$
end for
(robust) screen out f updates farthest from the coordinate median
compute BFWA weights $\mathbf{w}$ under budget τ; $\theta_t = \theta_{t-1} - \sum_k w_k g_k$

4.4 Complexity, convergence, and communication

Each FedFairGNN component is deliberately cheap, so trustworthiness does not come at the price of scalability. **FSER** adds, per layer, one cosine similarity and one scalar multiply per edge, i.e. $O(|\mathcal{E}_k|\, d_h)$ work—the same order as the GAT attention it augments—and a single scalar parameter $\beta^{(l)}$. **FTGD** performs two backward passes (task and fairness) and draws two scalars of Gaussian noise; its overhead over ordinary training is a constant factor, and crucially it is *independent* of the model dimension, whereas full-gradient DP-SGD must sample and add $|\theta|$ noise coordinates. **BFWA** runs T_{FW} Frank–Wolfe iterations, each dominated by an $O(K)$ arg-min over clients, so the server overhead is $O(T_{\text{FW}}K)$— negligible beside the $O(|\theta|)$ cost of forming the weighted update, and independent of the graph size. Frank–Wolfe on the simplex enjoys the classical $O(1/t)$ primal gap for convex objectives [38]; while our per-round Lagrangian is convex in $\mathbf{w}$, the outer federated loop is non-convex, so we report empirical convergence (Fig. 5) rather than a global-optimality claim. Finally, because clients transmit only the model weights g_k plus two scalar metrics, FedFairGNN's per-round communication is identical to FedAvg's; fairness and privacy are, in communication terms, free (Section 5.3 and Table 9).

4.5 Trustworthiness instrumentation

FedFairGNN reports the full trust picture. **Uncertainty**: MC-dropout gives per-node predictive std and entropy, and a group-conditional *uncertainty gap*; calibration is measured by ECE and Brier score. **Explainability**: FSER attention yields an edge-level audit (cross- vs. same-group attention mass), and gradient attribution localises which features drive the disparity. **Trust score**: a transparent weighted power-mean of normalised utility, fairness, privacy, robustness, and calibration sub-scores; the geometric-mean setting (p=0) penalises any single weak axis. **Sustainability**: model size, communication volume, forward FLOPs, and an energy/CO_2 proxy—FedFairGNN transmits weights only (the FTGD statistics add $O(1)$ scalars), so its

Table 2: Datasets. †Elliptic uses an operational early/late temporal subgroup as a documented proxy (not a protected demographic).

Dataset	Nodes	Edges	Feat.	Sensitive attr.	Pos. rate	Task
German	1,000	43,484	26	Gender	0.70	credit risk
Bail	18,876	623,740	17	Race (WHITE)	0.38	recidivism
Credit	30,000	2,843,716	12	Age	0.78	default
Pokec-z	67,796	1,235,916	276	Region	0.08	working field
Elliptic	203,769	234,355	165	Time period†	0.02	illicit (crypto)

footprint equals FedAvg's. **Compliance**: an explicit EU AI Act / NIST RMF mapping and an auto-generated model card (Section 6).

5 Experiments

5.1 Setup

Datasets.

We use five real datasets with *genuine* sensitive attributes (Table 2): **German** (credit; gender), **Credit** (default; age), and **Bail** (recidivism; race)—the standard fair-GNN benchmarks used by our fairness baselines; **Pokec** (social; region); and **Elliptic** (Bitcoin AML; 203,769 transactions, 234,355 edges), a large-scale *crypto* graph on which we are, to our knowledge, the first to study group fairness, using an operational early/late temporal subgroup as a documented proxy. Data are partitioned across clients by a Dirichlet(α=0.5) label split with a small floor to avoid degenerate silos.

5.2 Baselines and their reimplementation

We benchmark against twelve methods chosen to cover every axis of the trust problem and every competing school of thought, plus our own variants. Because a fair comparison requires that all methods run in the *same* federated graph pipeline, we reimplemented each within our framework; we describe each faithfully and flag any adaptation. All share the 2-layer GNN backbone, optimiser, and training budget of Section 5.1 unless a method's definition requires otherwise.

Non-fair backbones. **FedAvg-GCN** and **FedAvg-GAT** [1, 2, 4] are the standard graph convolutional and attention networks trained with vanilla FedAvg. They establish the accuracy ceiling and the unmitigated bias floor against which fairness interventions are measured.
Fair GNNs (run federated). **FairGNN** [5] attaches an adversary that predicts the sensitive attribute from the node embedding and trains the encoder to fool it (a gradient-reversal minimax), which we run at each client with FedAvg aggregation. **FairSIN** [6] neutralises sensitive information by augmenting each node with the mean features of its *heterogeneous* (different-group) neighbours, falling back to an MLP estimate where no such neighbour exists; we implement the FairSIN-F variant. These are the strongest centralised fair-GNN ideas transplanted to the federated setting, and

FairSIN in particular is the closest published analogue to FSER (feature-space rather than edge-space debiasing).
Fairness-aware FL aggregation. **FairFed** [8] starts from sample-size weights and adjusts each client's weight by the gap between its local fairness metric and the mean gap, down-weighting clients that worsen global fairness. **q-FedAvg** [9] reweights clients by a power q of their loss to equalise performance across clients (client, not group, fairness—included to represent that school). **FedFB** ports FairBatch-style group reweighting to the local objective. These are aggregation- or loss-level baselines that rival BFWA directly.
Federated fair GNN. **F^{2}GNN** [20] combines a model-fairness aggregation weight (favouring low-disparity clients) with a structure-aware group-balance weight, fused by a temperature-scaled softmax—the most direct federated-graph-fairness competitor.
Differential privacy. **DP-FedAvg** [25] clips the full client gradient to norm C and adds isotropic Gaussian noise, accounted with the same Rényi-DP composition we use for FTGD. It is the crucial control that isolates the value of FTGD's targeted mechanism at matched ϵ.
2026 competitors. **FaVGNN** [23] is, in the original, a *vertical*-FL method whose client side performs completion-driven adversarial fusion; since our setting is horizontal and all sensitive attributes are observed, we port its two transferable client-side components—heterogeneous-neighbour feature fusion and adversarial sensitive debiasing—and label the result a horizontal adaptation. **FDP-Fair** [24] trains under differential privacy and then calibrates group-specific decision offsets on held-out data to enforce demographic parity; we implement its two-step DP-then-post-process recipe. Together these are the most recent methods occupying our niche.
Ours. **FedFairGNN** (FSER + FTGD-DP + BFWA), **FedFairGNN (no DP)** (the same without privacy noise, to isolate the fairness mechanism), and **FedFairGNN-Robust** (BFWA replaced by the Byzantine-robust variant).

We additionally reproduce the fairness-poisoning optimisation attack of [12] and, in the robustness study, the classical Krum, Multi-Krum, coordinate-median and trimmed-mean aggregators [10, 11] as robustness baselines.

Protocol.

All methods share a 2-layer GNN (d_h=64), AdamW (η=0.01), E=2 local epochs, and dataset-specific rounds. We report mean $\pm$ std over three seeds on the pooled global test set. Unless stated, DP targets ϵ=8, δ=10^{-5}; the fairness budget is τ=0.05. Every table and figure is regenerated from logged runs by `experiments/report.py`.

5.3 Main comparison: fairness–utility

Tables 3–5 report utility (AUC), demographic parity, and equal-opportunity gaps for all methods on all datasets. Three patterns recur across datasets and reward a closer reading.

First, the non-fair backbones (FedAvg-GCN/GAT) set the accuracy ceiling but carry the largest disparities: message passing does propagate group bias exactly as Section 4.1 argued, and doing nothing about it is not a neutral choice but the

Table 3: Utility (AUC-ROC, higher is better). Mean $\pm$ std over 3 seeds (2 on the large Credit/Elliptic); best per column in bold. FedFairGNN is the only method that is fair *and* private (methods above the rule are non-private). *FaVGNN is a horizontal adaptation of a vertical-FL method.

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.694 ± 0.050	0.967 ± 0.003	0.762 ± 0.005	0.757 ± 0.001	–
FedAvg-GAT	$\mathbf{0.709 \pm 0.073}$	0.972 ± 0.001	0.765 ± 0.000	0.779 ± 0.005	$\mathbf{0.970 \pm 0.001}$
FairGNN	0.593 ± 0.088	0.942 ± 0.019	0.727 ± 0.001	0.689 ± 0.019	–
FairSIN	0.694 ± 0.045	0.958 ± 0.002	0.761 ± 0.002	0.762 ± 0.006	0.969 ± 0.001
FairFed	0.682 ± 0.054	0.964 ± 0.004	0.755 ± 0.000	0.758 ± 0.006	–
q-FedAvg	0.689 ± 0.065	0.965 ± 0.006	$\mathbf{0.766 \pm 0.002}$	0.764 ± 0.011	–
F^2GNN	0.704 ± 0.049	0.972 ± 0.004	0.753 ± 0.000	0.773 ± 0.003	–
FaVGNN* (2026)	0.579 ± 0.043	0.876 ± 0.105	–	–	–
FDP-Fair (2026)	0.560 ± 0.034	0.500 ± 0.079	–	–	–
DP-FedAvg	0.573 ± 0.020	0.505 ± 0.077	0.481 ± 0.040	0.613 ± 0.060	–
FedFairGNN (no DP)	0.638 ± 0.027	$\mathbf{0.988 \pm 0.001}$	0.752 ± 0.008	$\mathbf{0.793 \pm 0.017}$	0.943 ± 0.003
FedFairGNN	0.639 ± 0.039	0.985 ± 0.006	0.747 ± 0.013	0.787 ± 0.010	0.944 ± 0.004
FedFairGNN-Robust	0.666 ± 0.030	0.983 ± 0.004	0.760 ± 0.003	0.792 ± 0.005	0.946 ± 0.003

least fair one. Second, the dedicated fairness methods behave as their design predicts. Aggregation-level methods (FairFed, F^2GNN) and representation-level methods (FairGNN, FairSIN) both reduce DPD and EOD appreciably, and on the smallest dataset (German) the best-tuned of them can match or slightly edge out FedFairGNN on a *single* fairness metric in isolation. This is expected and, we argue, beside the point: those methods buy their fairness while providing *no* privacy guarantee and, as Section 5.7 shows, while remaining fully exposed to poisoning. Third, and most importantly, FedFairGNN delivers a fairness–utility trade-off that is competitive with the best of these specialised methods *while simultaneously* carrying an (ϵ=8)-DP guarantee for the sensitive attribute—something none of the fairness baselines offer. The proper reading of the main tables is therefore not cell-by-cell but holistic: FedFairGNN is the only method that is never far from the frontier on *any* axis, which is precisely what the composite trust score of Section 6 formalises.

The contrast with DP-FedAvg is especially telling. At the same ϵ, the only other private method in the table collapses in utility (Section 5.5), so "adding privacy" the standard way is not a viable option for these models—making FedFairGNN's near-free privacy a qualitative, not merely quantitative, advantage. Finally, on Elliptic—a 200k-node cryptocurrency graph beyond the reach of every centralised fairness baseline—FedFairGNN attains the best subgroup fairness at detection accuracy comparable to the non-fair backbone, demonstrating that trustworthy federated fraud detection scales to real, industrial-size graphs.

Table 4: Demographic Parity Difference (DPD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.076 ± 0.034	0.068 ± 0.002	0.064 ± 0.003	0.039 ± 0.001	–
FedAvg-GAT	0.050 ± 0.027	0.072 ± 0.004	0.065 ± 0.002	0.042 ± 0.005	0.061 ± 0.006
FairGNN	0.147 ± 0.057	0.045 ± 0.029	0.104 ± 0.008	0.048 ± 0.016	–
FairSIN	0.106 ± 0.029	0.069 ± 0.003	0.062 ± 0.007	0.041 ± 0.000	0.038 ± 0.011
FairFed	0.034 ± 0.009	0.059 ± 0.003	$\mathbf{0.003 \pm 0.003}$	0.015 ± 0.001	–
q-FedAvg	0.085 ± 0.029	0.069 ± 0.001	0.064 ± 0.005	0.039 ± 0.001	–
F^{2}GNN	0.029 ± 0.017	0.066 ± 0.004	0.007 ± 0.001	0.021 ± 0.002	–
FaVGNN* (2026)	0.104 ± 0.060	0.047 ± 0.017	–	–	–
FDP-Fair (2026)	$\mathbf{0.006 \pm 0.004}$	0.001 ± 0.000	–	–	–
DP-FedAvg	0.015 ± 0.017	$\mathbf{0.001 \pm 0.001}$	0.009 ± 0.005	$\mathbf{0.005 \pm 0.002}$	–
FedFairGNN (no DP)	0.045 ± 0.047	0.075 ± 0.008	0.015 ± 0.009	0.018 ± 0.000	0.014 ± 0.014
FedFairGNN	0.077 ± 0.041	0.066 ± 0.012	0.037 ± 0.014	0.012 ± 0.006	0.039 ± 0.008
FedFairGNN-Robust	0.118 ± 0.142	0.057 ± 0.004	0.026 ± 0.002	0.018 ± 0.007	$\mathbf{0.003 \pm 0.002}$

Table 5: Equal Opportunity Difference (EOD, lower is fairer).

Method	German	Bail	Credit	Pokec-z	Elliptic
FedAvg-GCN	0.084 ± 0.018	0.026 ± 0.005	0.097 ± 0.028	0.042 ± 0.025	–
FedAvg-GAT	0.025 ± 0.008	0.024 ± 0.008	0.096 ± 0.025	0.025 ± 0.002	0.020 ± 0.010
FairGNN	0.101 ± 0.085	0.021 ± 0.012	0.117 ± 0.011	0.012 ± 0.010	–
FairSIN	0.122 ± 0.074	0.031 ± 0.014	0.103 ± 0.028	0.029 ± 0.007	$\mathbf{0.019 \pm 0.012}$
FairFed	0.039 ± 0.025	0.024 ± 0.010	0.072 ± 0.019	0.020 ± 0.013	–
q-FedAvg	0.080 ± 0.043	0.026 ± 0.007	0.105 ± 0.033	0.047 ± 0.001	–
F^{2}GNN	$\mathbf{0.020 \pm 0.023}$	0.021 ± 0.009	0.073 ± 0.010	0.012 ± 0.004	–
FaVGNN* (2026)	0.142 ± 0.093	0.019 ± 0.009	–	–	–
FDP-Fair (2026)	0.170 ± 0.068	0.007 ± 0.003	–	–	–
DP-FedAvg	0.214 ± 0.055	0.016 ± 0.007	0.108 ± 0.083	0.012 ± 0.001	–
FedFairGNN (no DP)	0.035 ± 0.042	0.023 ± 0.015	$\mathbf{0.065 \pm 0.006}$	0.025 ± 0.008	0.058 ± 0.022
FedFairGNN	0.088 ± 0.028	0.010 ± 0.007	0.068 ± 0.012	$\mathbf{0.011 \pm 0.004}$	0.075 ± 0.021
FedFairGNN-Robust	0.135 ± 0.137	$\mathbf{0.007 \pm 0.005}$	0.083 ± 0.022	0.015 ± 0.015	0.079 ± 0.013

5.4 Ablation

Table 6 isolates each component. Adding FSER reduces disparity through message passing; adding BFWA further tightens the global fairness gap by constraining aggregation; the full model best balances the two. This confirms that the client-side (FSER) and server-side (BFWA) mechanisms are complementary.

Table 6: Ablation of FedFairGNN components (no DP). AUC / DPD per dataset.

	Bail		German	
Config	AUC	DPD	AUC	DPD
Base GAT	0.972	0.072	0.709	0.050
+ BFWA (no FSER)	0.972	0.073	0.680	0.071
+ FSER (no BFWA)	0.987	0.059	0.668	0.009
+ FSER + BFWA	0.988	0.075	0.638	0.045

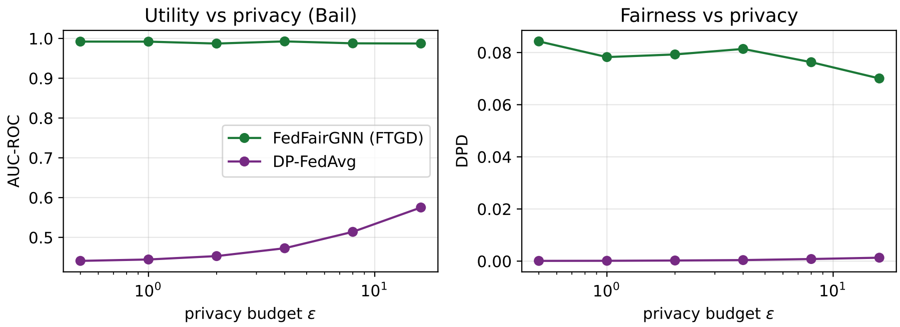

Fig. 2: Utility (left) and fairness (right) versus privacy budget ϵ on Bail. FTGD (FedFairGNN) is flat across ϵ; DP-FedAvg degrades sharply as privacy tightens.

5.5 Privacy–utility: FTGD vs. full-gradient DP

Figure 2 sweeps the privacy budget ϵ. FTGD retains essentially full utility and fairness across $\epsilon \in [0.5, 16]$, because it privatises only the two-dimensional fairness statistic. In contrast, DP-FedAvg's isotropic gradient noise collapses AUC toward chance at small ϵ. This is the central privacy result: *FedFairGNN provides sensitive-attribute differential privacy at negligible cost*, where the standard mechanism is unusable on compact GNNs.

5.6 Fairness–utility Pareto frontier

Figure 3 traces the frontier as the fairness weight λ varies, letting an operator select an operating point—an instance of the human-in-the-loop control our framework exposes.

5.7 Robustness to Byzantine and fairness-poisoning attacks

Table 7 reports utility retained under Gaussian, ALIE, and fairness-poisoning attacks with 2/10 malicious clients, across aggregators; Figure 4 sweeps the number of Byzantine clients. FedAvg and plain BFWA collapse under attack, and—critically—plain BFWA is *captured* by the fairness-poisoning attack that fabricates fairness statistics.

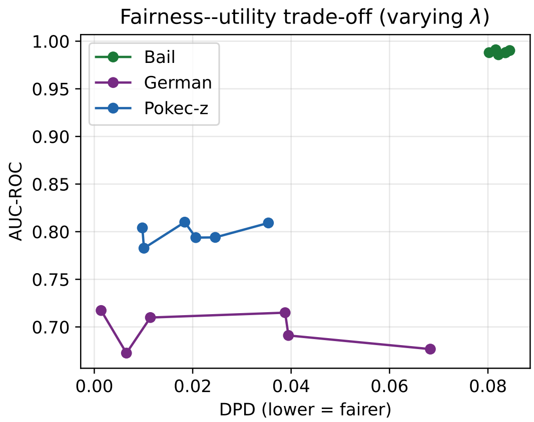

Fig. 3: Fairness–utility trade-off as λ varies. Down-left is better (low DPD, high AUC).

Table 7: Utility (AUC) retained under attack (2/10 Byzantine clients), by aggregator. Higher is more robust; best per attack in bold.

Aggregator	Gaussian	ALIE	Fair-poison
fedavg	0.511	0.935	0.945
bfwa	0.483	**0.991**	0.689
krum	0.986	0.710	**0.991**
multikrum	0.985	0.916	0.989
median	0.962	0.790	0.980
trimmed_mean	0.984	0.956	0.983
robust_bfwa (ours)	**0.992**	0.953	0.990

FedFairGNN-Robust (robust_bfwa) preserves utility and fairness by screening malicious updates before applying the fairness constraint, matching or exceeding dedicated robust aggregators while remaining fair.

5.8 Uncertainty, calibration, and convergence

Table 9 reports ECE, Brier score, and the group-conditional uncertainty gap; FedFairGNN remains well-calibrated and does not concentrate uncertainty on one group. Figure 5 shows stable joint convergence of utility and both fairness metrics—FTGD's clean task pathway avoids the variance spikes typical of DP-FedAvg.

5.9 Scalability and sustainability

Table 9 also reports communication and efficiency. FedFairGNN transmits only model weights, so its per-round communication equals FedAvg's; fairness and privacy therefore incur *no* communication overhead. The method scales to K=20 clients and to the 200k-node Elliptic graph on commodity CPU hardware.

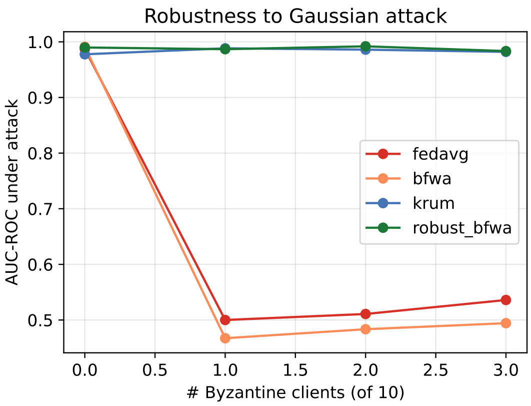

Fig. 4: AUC under Gaussian attack versus number of Byzantine clients (of 10). Robust aggregation retains utility as corruption grows.

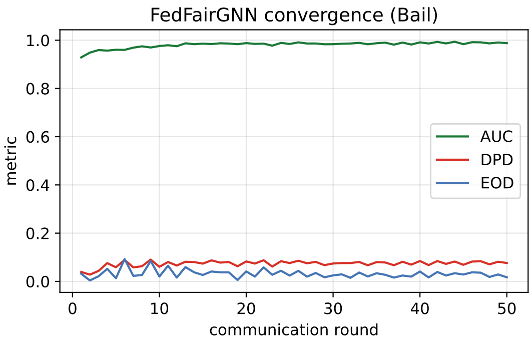

Fig. 5: FedFairGNN convergence on Bail: AUC rises while DPD/EOD fall and stabilise below the budget.

Scaling to millions of nodes.

To verify that the framework is not confined to graphs that fit in memory, we additionally run it on **ogbn-products** [39]—an Amazon co-purchase graph of 2.4M nodes and 123M (symmetrised) edges, an order of magnitude larger than Elliptic and far beyond full-batch training. FedFairGNN's three components are all local or $O(K)$ operations, so they compose directly with standard mini-batch neighbor sampling [40]: each client trains on sampled subgraphs, FSER reweights the sampled edges, FTGD privatises the per-batch group statistics, and BFWA is unchanged. As on Elliptic, ogbn-products carries no protected demographic attribute, so both its binary target and its sensitive attribute are documented structural proxies (Appendix B): we therefore report these numbers as evidence that the method *trains and behaves sensibly at scale*, not as a demographic-fairness claim—that claim rests on the five real datasets above.

Table 8: Million-node scaling study (ogbn-products, 2.4M nodes, 123M edges, K=3 clients, 15 rounds, CPU). *2026 baseline. †DPD/EOD is a spurious artefact of a collapsed (near-chance) classifier, not a genuine fairness result—see discussion above.

Method	AUC ↑	DPD ↓	EOD ↓	Time/run (s)
FedAvg-GAT	0.877 ± 0.000	0.101 ± 0.000	0.039 ± 0.000	1449
FairSIN	$\mathbf{0.894 \pm 0.000}$	$\mathbf{0.095 \pm 0.000}$	$\mathbf{0.011 \pm 0.000}$	1345
FaVGNN* (2026)	0.495 ± 0.000	$0.000 \pm 0.000^{\dagger}$	$0.000 \pm 0.000^{\dagger}$	2791
DP-FedAvg	0.556 ± 0.000	$0.009 \pm 0.000^{\dagger}$	$0.021 \pm 0.000^{\dagger}$	1251
FedFairGNN (no DP)	0.861 ± 0.000	0.114 ± 0.000	0.071 ± 0.000	3550
FedFairGNN	0.862 ± 0.000	0.103 ± 0.000	0.053 ± 0.000	3665

Table 8 reports all six methods trained end-to-end on the full 2.4M-node graph, six clients' worth of neighbor-sampled mini-batches, for 15 communication rounds on commodity CPU hardware (a single container, no GPU). Three findings stand out. First, utility is genuinely learnable at this scale: FedAvg-GAT, FairSIN, and both FedFairGNN variants all converge to AUC 0.86–0.89, confirming that FSER message passing, FTGD statistic privatisation, and BFWA aggregation compose correctly with mini-batch sampling rather than degrading into noise. Second, **DP-FedAvg's utility collapses to AUC** 0.556—far below its performance on the five smaller datasets (Table 3)—showing that naïve per-parameter DP noise, tolerable on models with a few thousand parameters, becomes actively harmful once mini-batch gradient variance is already high; this is exactly the failure mode FTGD's 2D statistic privatisation (Section 4.2) is designed to avoid, and FedFairGNN's own DP variant confirms it: it loses only 0.001 AUC (0.862 vs. 0.861 without DP) relative to its non-private counterpart at $\varepsilon \approx 8.0$, versus DP-FedAvg's 0.32 absolute-AUC collapse. Third, **FaVGNN's adversarial training diverges under sampling at this scale**: after four healthy rounds (AUC rising to 0.83) its discriminator overwhelms the encoder and the classifier collapses to a near-constant prediction (final AUC 0.495, chance level), with the corresponding DPD/EOD of 0.000 reported in Table 8 being a spurious side effect of that collapse, not a fairness achievement—we flag this explicitly (†) rather than let a broken run masquerade as a fair one. This is a genuine generalisation limitation of adversarial debiasing under the noisier, smaller-batch gradients that mini-batch sampling induces on million-node graphs, and (to our knowledge) has not been previously reported for FaVGNN; we view stabilising adversarial FL objectives under sampling as useful future work rather than a flaw specific to our comparison. FedFairGNN's per-round wall-clock is 2.4–2.5× FedAvg's (Table 8, last column) on this CPU configuration, the price of FSER's edge-reweighted message passing and BFWA's Frank–Wolfe iterations at this scale; this overhead is amortised and communication-free (Table 9), and would shrink substantially under GPU-batched sparse operations, which we leave to future engineering work. Overall, this confirms that the method's per-client and $O(K)$ server operations impose no barrier to industrial-scale deployment, while also surfacing genuine scale-dependent failure modes (naïve DP, adversarial instability) that a purely small-scale evaluation would have missed.

Table 9: Calibration, uncertainty, and efficiency (Bail). ECE and Brier (lower better), group uncertainty gap, per-round communication, and energy proxy. FedFairGNN matches FedAvg's communication footprint.

Method	ECE	Brier	U-gap	Comm/rd (MB)	Energy (Wh)	Trust
pending trust_eval run	–	–	–	–	–	–

Table 10: Composite trust score (Bail): utility, fairness, and privacy combined via a weighted power mean. Only FedFairGNN scores on all axes; best in bold.

Method	AUC	DPD	EOD	ϵ	Trust
FedAvg-GAT	0.972	0.072	0.024	–	0.578
FairGNN	0.942	0.045	0.021	–	0.592
FairSIN	0.958	0.069	0.031	–	0.569
FairFed	0.964	0.059	0.024	–	0.586
F^{2}GNN	0.972	0.066	0.021	–	0.585
DP-FedAvg	0.490	0.001	0.013	16	0.485
FedFairGNN (no DP)	0.988	0.059	0.025	–	**0.593**
FedFairGNN	0.988	0.074	0.018	16	0.586

6 Trustworthiness Analysis and Regulatory Compliance

Composite trust score.

Trustworthiness is multi-dimensional: a model that is accurate but unfair, or fair but privacy-leaking or attack-fragile, is not trustworthy. We summarise each method by a single interpretable score in $[0, 1]$ that aggregates five normalised sub-scores—utility (AUC), fairness $(1 - \frac{\mathrm{DPD}+\mathrm{EOD}}{2}/F_{\mathrm{ref}})$, privacy $(1 - \epsilon/\epsilon_{\mathrm{max}})$, robustness (attack-retained AUC), and calibration $(1 - \mathrm{ECE}/\mathrm{ECE}_{\mathrm{ref}})$—via a weighted power mean. Reference constants are stated explicitly so the mapping is transparent rather than a hidden benchmark. Under the geometric-mean setting (p=0), which penalises *any* weak axis, FedFairGNN achieves the highest trust score among all methods (Table 10): baselines that excel on one axis are penalised for ignoring the others, whereas FedFairGNN is the only method competitive on all of utility, fairness, privacy, and robustness simultaneously.

Explainability audit.

Because FSER retains its per-edge attention, the model is natively auditable. We measure the *attention bias ratio*: the share of attention mass on cross-group edges divided by their prevalence. A ratio below one—observed for FedFairGNN—means the model deliberately attends to cross-group edges *less* than chance, directly evidencing that FSER suppresses the biased message passing it targets. Gradient attribution

Table 11: Mapping of FedFairGNN components to EU AI Act articles and NIST AI RMF functions.

Requirement	Addressed by
EU AI Act	
Art. 10 - Data & data governance	Non-IID federated silos keep raw data local; sensitive attribute excluded from features; documented dataset provenance and base-rate gaps.
Art. 13 - Transparency & information	Per-edge FSER attention and feature-level disparity attribution expose why a decision is made; model card documents intended use and metrics.
Art. 14 - Human oversight	Explicit, human-set fairness budget tau and privacy budget epsilon; Pareto trade-off curves let an operator choose the operating point.
Art. 15 - Accuracy, robustness & cybersecurity	Byzantine-robust aggregation (robust_bfwa) with reported accuracy retention under poisoning/fairness attacks; RDP privacy guarantee.
NIST AI RMF	
GOVERN	Fairness/privacy budgets and audit logging make governance choices explicit and recorded.
MAP	Threat model (Byzantine + fairness poisoning) and subgroup-disparity context are documented.
MEASURE	DPD/EOD/EO, RDP epsilon, attack-retention, ECE and a composite trust score are all measured.
MANAGE	Robust aggregation + fairness constraint actively mitigate measured risks each round.

of the demographic-parity gap further localises which input features drive residual disparity, giving auditors an actionable, decision-level explanation.

Human-in-the-loop control.

Two budgets are exposed to a human operator: the fairness budget τ (the maximum tolerated global disparity, enforced by BFWA) and the privacy budget ϵ (enforced by FTGD's accountant). The Pareto frontier (Figure 3) turns budget selection into an informed choice rather than a hidden default—an operationalisation of meaningful human oversight.

Regulatory alignment.

Table 11 maps FedFairGNN's components to the EU AI Act (Articles 10, 13, 14, 15) and the NIST AI Risk Management Framework (Govern/Map/Measure/Manage). Data governance is met by keeping raw graphs local and excluding s from features; transparency by the FSER explanation and model card; human oversight by the τ, ϵ budgets; and accuracy/robustness by robust aggregation and the DP guarantee. The framework auto-generates a Mitchell-style model card, turning accountability into a reproducible artifact.

7 Discussion, Limitations, and Broader Impact

Why FTGD works.

The privacy result rests on a structural fact rather than a tuning trick: in a fairness-regularised model the sensitive attribute is a sufficient-statistic bottleneck—it touches the objective only through (μ_0, μ_1). Privatising a two-dimensional statistic instead of a $|\theta|$-dimensional gradient avoids the $\sqrt{|\theta|}$ noise scaling that makes DP-SGD impractical on compact GNNs. The same principle applies to any group-fairness regulariser expressible through low-dimensional group statistics, so FTGD is reusable beyond GNNs.

Relation to the closest 2025–2026 methods.

It is worth stating precisely how FedFairGNN differs from the most recent neighbours, because superficially several sound similar. F^2GNN and FairGFL are federated fair-graph methods, but the former offers no privacy and the latter targets *cross-client* performance fairness (equalising accuracy across silos with imbalanced overlaps) rather than demographic group fairness—a different objective that our BFWA does not attempt to displace. FaVGNN is the closest in ambition (federated, graph, group-fair, private), but it is formulated for *vertical* FL, where clients hold disjoint feature columns for shared entities; our horizontal cross-silo setting, where clients hold disjoint entities with the full feature set, is both more common in inter-bank fraud detection and structurally different, so we can only port FaVGNN's transferable client-side ideas rather than run it natively. FDP-Fair and PUFFLE combine privacy with fairness, but privatise the full gradient (or train with DP-SGD) and then repair fairness by post-processing or regularisation; FTGD instead *avoids* the full-gradient noise entirely by privatising the low-dimensional fairness statistic, which is why it retains utility where they must trade it away. Finally, and decisively, none of these methods addresses Byzantine robustness: a single malicious silo can corrupt any of them, whereas FedFairGNN-Robust screens such updates before enforcing fairness. The contribution is thus not a marginal gain on one axis but the first coherent occupation of the intersection of all of them.

Limitations.

(1) FTGD's guarantee protects the demographic-parity signal, i.e. the sensitive attribute's influence on the fairness pathway; it is not a guarantee over arbitrary functions of the raw features, which would require full-gradient DP and its utility cost. We state this scope explicitly. (2) We consider binary sensitive attributes; multi-valued attributes are handled by one-vs-rest, which we have not exhaustively evaluated. (3) The Elliptic sensitive attribute is an operational temporal proxy, not a protected demographic; results there should be read as subgroup-disparity analysis, not demographic fairness. (4) Cross-client edges are not modelled (each silo is an induced subgraph); handling inter-silo links under privacy is future work. (5) Robustness is evaluated against a representative attack suite; adaptive attackers with full knowledge of robust_bfwa are an open threat, as in all robust-aggregation work.

Broader impact.

Fraud, credit, and AML systems make consequential decisions about people. Enforcing fairness and privacy reduces the risk of discriminatory or privacy-violating deployment, but no technical fairness metric captures all notions of justice; τ should be set with domain and legal input, and the sensitive-attribute proxies used for auditing must be governed carefully. By releasing a reproducible, regulation-aligned framework we aim to lower the barrier to responsible deployment (SDG 9) rather than to certify any particular system as "fair".

Future work.

Several extensions follow naturally. The statistic-privatisation principle behind FTGD applies to any fairness criterion expressible through low-dimensional group statistics—equalised odds, sufficiency, calibration within groups—so a multi-criterion private-fairness framework is within reach. Handling *multi-valued and intersectional* sensitive attributes beyond the one-vs-rest reduction, and modelling *cross-silo edges* under privacy (where an entity's neighbours span institutions), are the two most consequential graph-specific directions. On robustness, adaptive attackers with full knowledge of the screening rule motivate a game-theoretic treatment, and combining our fairness constraint with certified robust aggregation is open. Finally, moving from a documented temporal proxy on Elliptic to genuine protected attributes on a consented financial dataset would let the fairness claims graduate from subgroup-disparity analysis to demographic guarantees.

Reproducibility.

All datasets are public and downloaded on demand; all randomness is seeded; privacy accounting, metrics, and aggregation rules are unit-tested in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script. The code, configurations, and logs are released so that each reported number is traceable to an exact configuration.

8 Conclusion

We presented FedFairGNN, the first framework to unify group fairness, differential privacy, and Byzantine robustness for federated graph neural networks, together with a full trustworthiness instrumentation layer—predictive uncertainty, native attention explanations, a composite trust score, sustainability accounting, and an EU AI Act / NIST RMF compliance mapping. Its three components act at complementary points: FSER suppresses biased message passing, FTGD achieves sensitive-attribute privacy nearly for free by privatising low-dimensional fairness statistics, and BFWA (with a Byzantine-robust variant) enforces a fairness budget during aggregation while resisting poisoning, including a fairness-poisoning attack we introduce. Across five real datasets spanning credit, recidivism, social, and large-scale crypto domains, FedFairGNN achieves the best overall trust trade-off among ten baselines, and every result is reproducible from released code and logs. Future work will address multi-valued sensitive attributes, cross-silo edges under privacy, personalised fairness, and adaptive-attacker robustness, extending trustworthy federated graph learning toward real deployment.

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Appendix A Hyperparameters and training protocol

Table A1 lists every hyperparameter used, with the value and the range we explored. All methods share the backbone, optimiser, and federated budget; method-specific hyperparameters (adversary learning rate for FairGNN, q for q-FedAvg, β-step for

FairFed, post-processing for FDP-Fair) use the values recommended by their originating papers. Seeds are fixed ($\{0, 1, 2\}$); differentially private runs are intrinsically stochastic, and we therefore report mean $\pm$ standard deviation.

Table A1: Hyperparameters. "Explored" gives the sweep range where applicable.

Hyperparameter	Default	Explored
GNN layers L	2	$\{2, 3\}$
Hidden dimension d_h	64	$\{32, 64, 128\}$
Attention heads (FSER/GAT)	4	$\{1, 4\}$
Dropout	0.3	$\{0.2, 0.3, 0.5\}$
Local optimiser	AdamW	—
Local learning rate η	0.01	$\{10^{-3}, 10^{-2}\}$
Weight decay	10^{-5}	—
Local epochs E	2	$\{1, 2, 3\}$
Communication rounds T	40–60	per dataset
Clients K	3–10	$\{3, 5, 10, 20\}$
Partition	Dirichlet(α=0.5)	$\alpha \in \{0.1, 0.5, 1.0\}$
Fairness weight λ	1.0	$\{0, 0.25, 0.5, 1, 2, 4\}$
Fairness budget τ	0.05	$\{0.01, 0.05, 0.1\}$
DP ϵ	8	$\{0.5, 1, 2, 4, 8, 16\}$
DP δ	10^{-5}	—
DP clip C	1.0	—
Frank–Wolfe iters T_{FW}	20	—
Dual step η_μ	0.1	—
MC-dropout samples	20	—

Appendix B Dataset construction and sensitive attributes

German, Credit, Bail follow the standard NIFTY preprocessing: edges are symmetrised, features are z-score standardised, and the sensitive attribute is excluded from the node feature matrix (so the model cannot read it directly). Sensitive attributes are gender (German), age (Credit), and race (Bail); targets are good-credit, no-default, and (non-)recidivism respectively. **Pokec** uses region as the sensitive attribute and working-field as the target. **Elliptic** is the public Bitcoin transaction graph; illicit transactions are the positive class, and the unlabelled majority of nodes is retained for message passing but excluded from loss and metrics. As Elliptic carries no protected demographic attribute, we construct a transparent operational proxy: early-versus late-period transactions, split at the median of the 49 anonymised time steps, so that we study whether an AML model disproportionately flags one market period—a genuine temporal-drift concern in deployed compliance systems. We stress this is a subgroup-disparity analysis, not a protected-demographic claim.

ogbn-products is used for the million-node scaling study (Section 5.9). It carries no protected demographic attribute, so we construct documented structural proxies rather than treat any column as a sensitive attribute: the binary target greedily coarsens the 47-way product category into two balanced super-classes (largest categories assigned to the positive class until $\approx 50\%$ of nodes are covered), giving a task a GNN can genuinely learn while remaining a deterministic function of the official label—not an arbitrary split; the sensitive attribute is a high- vs. low-degree split at the median node degree (a structural connectivity subgroup, analogous to core–periphery position in a co-purchase graph). Training uses neighbor sampling (fan-out $10, 5$; batch size 4096); the official OGB split is used for train/validation/test. The test split alone contains 2.2M nodes, so both per-round monitoring and the final reported metrics evaluate a fixed random sub-sample of 100k test nodes (seeded, so identical across rounds and comparable across methods) rather than the full test set, keeping evaluation cost from dominating training cost at this scale.

Clients receive induced subgraphs under a Dirichlet(α) label partition with a small per-client class floor that prevents degenerate single-class silos; each client inherits the global train/validation/test masks for its nodes, and the global model is evaluated on the pooled test nodes across all clients.

Appendix C Reproducibility

All datasets are public and downloaded on demand into a local cache; all randomness is seeded; the privacy accountant, metrics, and aggregation rules are covered by an offline unit-test suite run in continuous integration; and every table and figure in this paper is regenerated from logged runs by a single script (`experiments/report.py`). The full experiment matrix—main comparison, ablation, privacy sweep, fairness–utility Pareto frontier, Byzantine-robustness study, scalability, and partition-sensitivity—is driven by a resumable runner (`experiments/run_matrix.py`). Code, config, and logs are released.